

Group-Agent Reinforcement Learning for Stock Trading

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Abstract

This project investigates whether Group-Agent Reinforcement Learning (GARL) can improve stock trading performance by allowing agents assigned to separate equities to exchange learning knowledge asynchronously. Two financial adaptations of GARL-DDAL are evaluated against passive allocation, statistical forecasting, traditional supervised ML, deep sequence models, as well as joint and independent implementations of A2C, PPO and DDQN. The experiment uses daily data for nine US-listed equities from 2001 to 2025 within a causal and cost-aware framework. Models are developed through five chronological folds and tested on a final holdout period from January 2023 to December 2025. All model-generated positions incorporate execution delay and trading frictions, while RL results are averaged across ten independent training seeds.

Random Forest achieved the highest holdout Sharpe ratio of 2.059 under low market exposure, whereas equal-weight buy-and-hold produced the highest cumulative return of 228.1%. Among RL strategies, independent A2C achieved a mean Sharpe ratio of 1.835 and cumulative return of 128%. GARL-DDAL produced closely comparable results, with a Sharpe ratio of 1.833 and cumulative return of 124.9%, while maintaining lower exposure and drawdown than buy-and-hold. The selective GARL-DDAL extension reduced peer sharing and rejected conflicting gradients as intended, but did not consistently improve aggregate performance.

Across development folds, passive allocation and stock-specific actor-critic strategies formed the leading group. The findings show that asynchronous gradient transfer can preserve competitive stock-specific policies, but its benefit depends on the economic relevance of the exchanged knowledge. This study contributes a controlled characterisation of GARL in a non-stationary financial setting and identifies regime-aware transfer selection as a promising future work direction for collaborative trading agents.

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1 Introduction

1.1 Background

Financial markets are institutions through which claims on future cash flows and risks are issued, exchanged and valued. Common tradable instruments include equities, bonds, foreign exchange, commodities and derivatives. This project focuses on equities, in which a share represents a fractional ownership claim in a company. Stock trading changes that ownership exposure in response to expected return, risk or liquidity needs. A trading strategy must determine which portfolio assets to hold, the direction and magnitude of each position, and when to change those positions.

These decisions are challenging because daily returns are noisy, their distributions and cross-asset relationships vary over time, and even a predictive signal can be unprofitable after transaction costs. The Efficient Market Hypothesis (EMH) [1] formalises part of the difficulty. Its weak, semi-strong and strong forms vary according to whether prices incorporate past market data, public-only information, or all available public and private information, respectively. The hypothesis does not imply that prices are always equal to an observable true value, nor that stock price prediction must always fail. Rather, it establishes a demanding benchmark that an apparent historical pattern is not sufficient evidence of an exploitable strategy. Lo's Adaptive Market Hypothesis (AMH) [2] further suggests that market efficiency may vary as participants, competition, and conditions change. For empirical trading research, the central question is consequently not only whether a model can fit historic returns, but whether it retains out-of-sample economic value across market regimes and after realistic costs.

Trading approaches are broadly categorised as discretionary or systematic. Discretionary trading relies primarily on human judgement, while systematic trading applies reproducible decision rules that may be deterministic or stochastic. Within this domain, quantitative trading denotes the application of mathematical, statistical, and computational models to formulate these rules, while algorithmic trading refers to their automated execution. Although these concepts frequently intersect, they are neither synonymous nor strictly hierarchical.

In systematic equity trading, classifying strategies by the model's functional role provides greater analytical clarity than treating quantitative, algorithmic, and machine learning methods as mutually exclusive categories. Under this framework, passive rules determine asset allocations without generating explicit return forecasts. Forecast-first approaches estimate future asset returns or directional probabilities and subsequently map those expectations to portfolio weights. In contrast, reinforcement learning bypasses an explicit return forecast by directly optimising a policy that maps market and portfolio states to actions with the objective of maximising expected cumulative cost-adjusted

reward. Fig. 1 places the methods evaluated in this report within this decision-theoretic taxonomy.

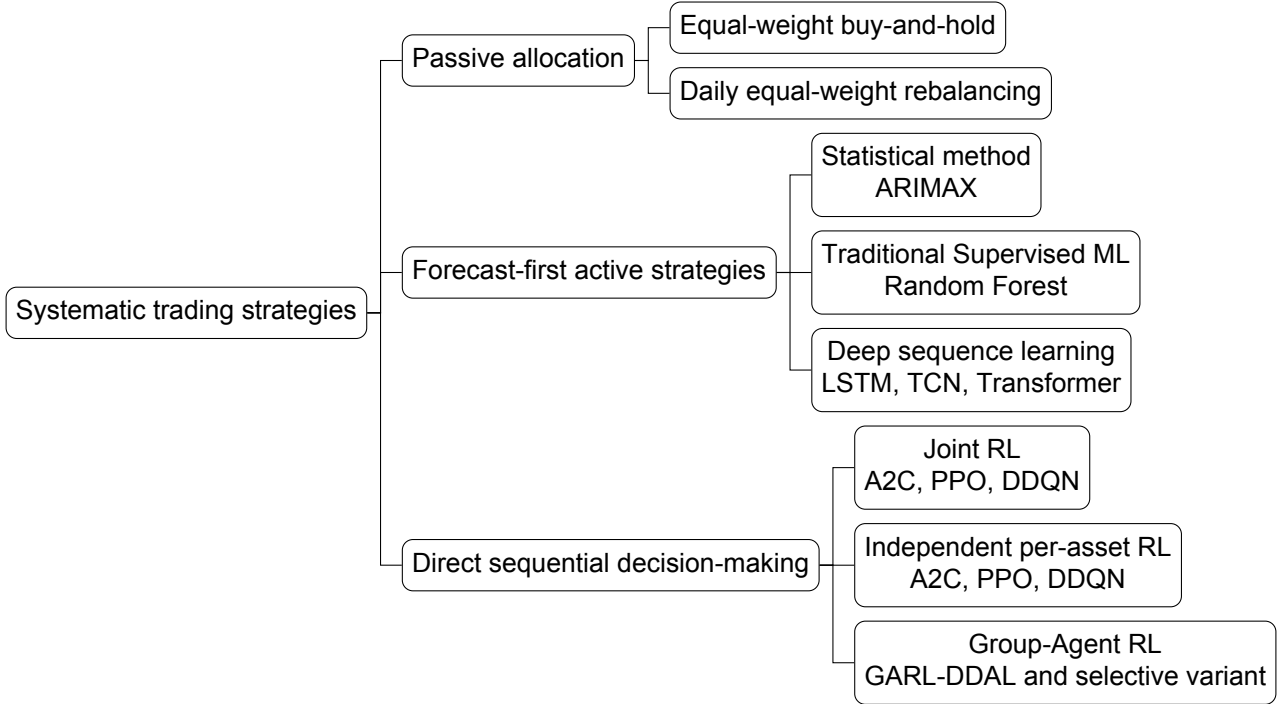


Fig. 1. Systematic trading strategies evaluated in this report.

The distinction between return prediction and portfolio decision-making is important because high accuracy in return prediction does not necessarily translate into profitable trading outcomes. A moderate but stable signal may retain economic value when exposure and turnover are controlled, whereas a volatile forecast may be exhausted by transaction costs. Therefore, this project evaluates complete trading strategies within an end-to-end performance framework, rather than forecasting loss alone. Passive allocation strategies follow predefined allocation rules. The forecast-first strategies translate predicted returns into bounded positions. RL strategies learn actions directly from market and portfolio states. Their detailed target definitions, decision frequency, execution delay, exposure constraints and transaction cost assumptions are specified in Section 2.

1.2 Literature Review

1.2.1 Market predictability and empirical evaluation

Research on return predictability regularly lies between two extremes by either assuming that historical data contain no useful information or assuming that any in-sample regularity is actionable. Empirical assessments of technical trading rules have produced mixed results across different markets, periods and testing procedures, with profitability often weakened by costs and data-snooping controls [3]. Flexible machine learning models intensify this concern because model choice, hyperparameter search and repeated inspection of a backtest can select a strategy that fits market noise.

Bailey et al. [4] formalised this as the probability of backtest overfitting, while Henderson et al. [5] showed that stochastic Deep RL results can vary materially with implementation and random seed.

These findings motivate three core requirements for the comparative framework. First, training, model selection, and testing must strictly respect chronological order. Second, all active strategies must share the same execution and friction assumptions, while passive benchmarks retain their defining allocation rules. Third, stochastic RL methods must be evaluated across multiple independent random-seed repetitions rather than represented by a single favourable run. The five development folds and the final holdout period used in this study are designed to expose temporal instability, while ten independent trials per RL baseline capture training variance. This evaluation methodology is as critical to this study as model development itself, since without rigorous assessment, greater architectural capacity is easily misinterpreted as delivering genuine economic value.

1.2.2 Statistical Forecasting

Statistical time-series models offer an interpretable benchmark against which more expressive architectures can be evaluated. The ARIMA model represents a differenced series using autoregressive and moving average terms [6], while the ARIMAX model augments this structure with exogenous predictors. For a differenced series $y'_t = (1 - B)^d y_t$, the ARIMAX(p, d, q) specification is formulated as:

$$y'_t = c + \sum_{i=1}^p \phi_i y'_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \beta^\top \mathbf{x}_t + \epsilon_t \quad (1)$$

where p , d and q denote the autoregressive, differencing and moving average orders, respectively, while $\mathbf{x}_t$ contains exogenous technical predictors. ARIMAX provides a transparent and computationally inexpensive benchmark, but its fixed linear structure could fail to capture non-linear feature interactions or changing market relationships. ARIMA-based models have previously been applied to stock price prediction [7], supporting their use as a familiar statistical reference. This research evaluates both static and rolling ARIMAX variants. The static model keeps its fitted parameters throughout evaluation, whereas the rolling model is periodically re-estimated using only observations whose five-day target horizons have fully elapsed.

1.2.3 Traditional Supervised Machine Learning

Supervised machine learning estimates a mapping $f : \mathbf{x}_t \mapsto \hat{r}_t^{(5)}$ from features observed at time t to the subsequent cumulative five-day trading return. Random Forest provides a non-linear supervised learning benchmark by averaging decision trees fitted to bootstrap samples and random feature subsets [8]. This construction can represent non-linear interactions while reducing the variance of an

individual tree. Krauss et al. applied Random Forest to statistical arbitrage on S&P 500 constituents [9], demonstrating the method’s relevance to this study. However, Random Forest does not model temporal order, so the chronology must be represented through causal indicators and lagged features. It also predicts returns instead of portfolio decisions, so its forecasts are converted into positions using the common allocation rule defined in Section 2.

1.2.4 Deep Sequence Learning

Deep sequence models learn temporal representations from ordered windows of market observations. Long Short-Term Memory (LSTM) networks use gated recurrent state updates to retain relevant historical information [10] and have been utilised for market movement prediction [11]. Temporal Convolutional Networks (TCNs) address the sequential dependency inherent in LSTM models by replacing recurrent mechanisms with causal dilated convolutions and residual connections. This approach provides a finite receptive field and enables parallel computation [12]. TCN-based architectures have also been implemented for financial price change prediction [13]. The Transformer architecture introduced the multi-head attention mechanism and positional encoding as alternatives to recurrent and convolutional sequence processing [14]. Its encoder, when combined with a causal attention mask, has demonstrated effectiveness in stock movement prediction [15]. Collectively, the LSTM, TCN, and Transformer encoder embody three distinct inductive biases: gated recurrence, causal convolution, and causal masked self-attention. Comparing these models enables evaluation of their performance under standardised inputs, prediction targets, and backtesting assumptions.

1.2.5 Deep Reinforcement Learning and Agent Organisation

The preceding forecast-first architectures treat trading as a supervised regression task, mapping historical observations to expected returns before applying a heuristic rule to determine portfolio weights. In contrast, Reinforcement Learning (RL) formulates trading directly as a Markov Decision Process (MDP) for sequential decision-making. At each discrete time step t , an agent observes an environmental state s_t , selects an action a_t according to a policy $\pi(a_t|s_t)$, receives an immediate scalar reward r_{t+1} , and transitions to the subsequent state s_{t+1} . The optimisation objective maximises the expected cumulative discounted return:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t r_{t+1} \right] \quad (2)$$

where $\gamma \in [0, 1]$ is the discounting factor that controls the trade-off between present and future rewards [16]. The reward function can directly incorporate execution frictions, portfolio inertia, and transaction costs into the optimisation objective. Moody and Saffell established early foundations in financial RL

with their direct reinforcement framework [17]. Later research has increasingly examined modern policy-gradient architectures, particularly actor-critic algorithms such as A2C and PPO [18]. Consequently, this work assesses these policy-based methods alongside the value-based DQN family to provide a balanced comparison across distinct reinforcement learning paradigms.

DQN [19] approximates the state-action value function $Q(s, a; \theta)$, from which a greedy policy is obtained by selecting the action with the highest estimated value. Standard DQN is susceptible to maximisation bias because it uses the same network parameters to select and evaluate candidate actions over noisy Q-value estimates. Double DQN (DDQN) addresses this overestimation by decoupling the action selection mechanism from the evaluation step [20]. Specifically, the online network determines the greedy action, while the periodically updated target network evaluates that action. Although extensions such as Rainbow DQN [21] incorporate multiple architectural enhancements, the increased complexity of full Rainbow configurations complicates interpretation and impedes clear attribution of performance improvements. Consequently, this study adopts only DDQN for value-based RL.

Whereas DDQN represents its policy through action values, actor-critic methods parameterise a policy $\pi(a | s; \theta_\pi)$, while learning a critic $V^\pi(s; \theta_v)$. The advantage estimate indicates how an action performed relative to the critic's expectation and reduces gradient variance compared to a vanilla policy-gradient estimator. Mnih et al. introduced Asynchronous Advantage Actor-Critic (A3C) [22], in which multiple workers interact with independent environment copies and asynchronously apply gradients to a shared parameter set without coordination. Although this parallelism decorrelates experience, gradients might be calculated from parameters that change before the update is applied. A2C replaces asynchronous application with synchronous updates from fixed rollouts. This avoids A3C's stale-gradient behaviour and facilitates controlled seed-based evaluation. Neither A3C nor A2C constrains the magnitude of policy changes. As a result, noisy advantage estimates may produce destabilising updates, which motivates successor algorithms to constrain update sizes.

Schulman et al. introduced PPO, which limits policy changes through a clipped surrogate objective [23]. Unlike TRPO, which enforces a KL-divergence constraint using computationally intensive second-order optimisation [24], PPO achieves an approximate trust-region effect using first-order gradient optimisation:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (3)$$

where $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ represents the probability ratio between the updated and reference policies, $\hat{A}_t$ is the empirical advantage estimate, and $\epsilon \in (0, 1)$ defines the clipping threshold. In financial trading, PPO has emerged as a conservative policy optimisation algorithm. Its clipped surrogate objective can improve optimisation stability in a non-stationary financial setting by limiting abrupt policy

changes. However, PPO does not directly address distribution shift, limit portfolio turnover, or ensure stable out-of-sample performance.

Algorithm choice represents only one facet of an RL trading architecture, as the structural organisation of agents is equally important. A centralised joint agent can process all selected assets simultaneously and output a portfolio-level action vector, enabling cross-asset conditioning at the cost of a large state and joint action space. In contrast, independent agents decompose the portfolio into single-asset sub-problems to enable localised specialisation, though they sacrifice the capacity to transfer structural representations across equities. Group-Agent Reinforcement Learning (GARL) provides an intermediate paradigm. Instead of forcing agents to compete or cooperate within a single shared environment as in standard multi-agent RL, GARL deploys agents across isolated environments on dedicated tasks while facilitating knowledge exchange among them [25]. Within the paradigm, Wu and Zeng introduced the Decentralised Distributed Asynchronous Learning (DDAL) framework, allowing local agents to optimise individual policies independently before asynchronous weighted gradient aggregation across related market tasks.

The financial GARL-DDAL adaptation evaluated in this study assigns one A2C agent to each asset. All agents are initialised from a common parameter template but subsequently maintain independent model copies, optimisers, stock environments and action sampling streams. After an initial private learning phase, each agent continues local optimisation. At scheduled sharing events, the receiving agent combines its current local gradient with the latest available gradients from a limited number of peer agents. Asynchrony is represented through a deterministic event-driven simulation of agents progressing at heterogeneous local rates. The resulting implementation preserves the principal cross-task gradient-sharing concept of DDAL while customising its learner, state representation, relevance measure, and execution environment to financial trading. In the standard formulation, the mixing weight ω_{ij} assigned to a peer gradient g_j is governed by both source experience e_j and pairwise task relevance q_{ij} :

$$\omega_{ij} = \frac{1}{2} \frac{e_j}{\sum_{k \in \mathcal{P}_i} e_k} + \frac{1}{2} \frac{q_{ij}}{\sum_{k \in \mathcal{P}_i} q_{ik}}, \quad \tilde{g}_i = \sum_{j \in \mathcal{P}_i} \omega_{ij} g_j \quad (4)$$

where $\mathcal{P}_i$ denotes the peer candidate set comprising the recipient agent and its selected peers. In this baseline setup, peer relevance q_{ij} corresponds to the absolute return correlation during training. While treating strong inverse dependencies as informative, this absolute metric can transmit parameter updates that directly conflict with the recipient agent’s objective.

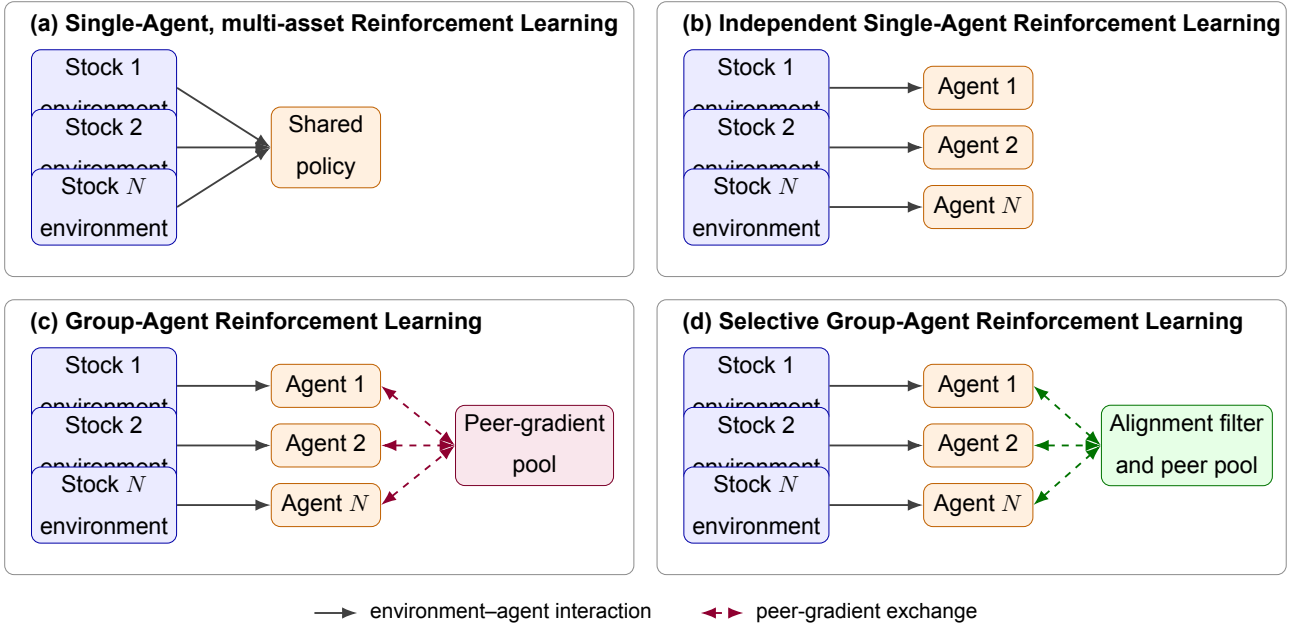


Fig. 2. Conceptual agent organisation of the RL strategies.

To address the risk of negative transfer, this project proposes Selective GARL-DDAL as an application-specific extension. This enforces dual filtering criteria that require a positive return correlation and a positive cosine gradient alignment between candidate peers and the recipient. Valid peer updates are further blended with the recipient’s local gradient. When no external candidate satisfies these compatibility constraints, the agent defaults entirely to an autonomous local update.

1.3 Research Gap and Motivation

Previous research has shown that statistical, ensemble, deep sequence, and reinforcement learning models can achieve strong forecasting or trading performance in financial domains. However, drawing direct conclusions across these studies is difficult because existing examinations often use different assets, target formulations, testing periods, transaction costs, and model-tuning budgets. Furthermore, in deep reinforcement learning studies, each independent baseline trial is highly sensitive to random-seed initialisation and trajectory stochasticity. Consequently, reported performance differences may partly reflect experimental design options rather than true algorithmic superiority.

A further important concern is how reinforcement learning agents are organised across related yet heterogeneous assets. Centralised joint-agent formulations risk over-generalisation by requiring the entire asset universe to be learned within a single parameter space, which may hinder adaptation to each asset’s dynamic characteristics. In contrast, independent single-asset agents prevent related peers from sharing beneficial experiences. The GARL approach offers a principled intermediate solution by allowing task-specific agents to exchange knowledge asynchronously, although its initial

validation was carried out on non-financial reinforcement learning benchmarks. Furthermore, equity returns introduce many challenges, such as low signal-to-noise levels, non-stationarity, regime-dependent shifts in cross-stock correlations, and execution costs. Thus, it remains an open empirical question whether asynchronous gradient transfer enhances financial decision-making, under which market conditions it may cause negative transfer, and whether receiver-side filtering mechanisms are able to successfully mitigate gradient interference.

This study tackles both concerns by implementing a standardised, end-to-end empirical framework. The framework benchmarks GARL baselines against passive baselines, predictive supervised architectures, and distinct reinforcement learning algorithms on conventional agent topologies, all under identical lookahead-free protocols and financial frictions. Both standard and selective GARL-DDAL variants are evaluated directly against independent A2C using matched training steps, seed repetition and tuning budgets. Importantly, empirical underperformance is not treated as a negative artefact but is instead analysed as evidence to delineate the operational boundaries and failure modes of collaborative group learning in financial markets.

1.4 Project Aim and Research Questions

The overarching aim of this research project is to critically evaluate whether, and under what conditions, asynchronous knowledge transfer among stock-specific RL agents improves, matches, or degrades out-of-sample trading performance relative to isolated learning, conventional agent organisations, forecast-based strategies, and passive investment benchmarks.

This study is structured around three core research questions:

- RQ1** How do statistical baselines, traditional supervised ML models, deep sequence architectures, and RL strategies compare against passive benchmarks when evaluated under a standardised, cost-aware, chronological validation framework?
- RQ2** How do RL algorithm choice and agent organisation influence realised returns, downside risk, turnover and training stability across random seeds?
- RQ3** Does asynchronous gradient transfer improve upon isolated single-agent learning, and can receiver-side knowledge selection reduce the impact of negative transfer in GARL?

1.5 Project Objectives

To address the research questions, eight technical and methodological objectives are defined:

1. **Causal Data Pipeline:** Construct a lookahead-free daily dataset for nine US-listed equities using adjusted OHLCV data, technical indicators, as well as a cumulative five-day return target.
2. **Benchmarking Infrastructure:** Implement a unified trading framework that includes equal-weight buy-and-hold, daily equal-weight rebalancing, static and rolling ARIMAX, Random Forest, LSTM, TCN, Transformer encoder, A2C, PPO, and DDQN baselines.
3. **RL Algorithm and Topology Evaluation:** Compare A2C, PPO, and DDQN under centralised joint and decentralised independent-agent organisations, employing a common feature schema, incremental action semantics, reward formulation, and an active-strategy execution model.
4. **Financial GARL Adaptation:** Adapt DDAL for multi-asset trading by assigning a dedicated A2C agent to each equity and incorporating common parameter initialisation, a private learning period, asynchronous gradient queues, and periodic weighted gradient aggregation.
5. **Selective Transfer Extension:** Develop a receiver-side selection method that integrates positive signed return relevance with cosine gradient alignment to mitigate possibly harmful peer transfer.
6. **Chronological Validation:** Implement five temporal development folds and a separate final holdout, with purging, hyperparameter selection, and feature scaling restricted to information available before each evaluation period.
7. **Performance and Stability Evaluation:** Compare strategies using return, volatility, downside risk, drawdown, turnover, exposure, and transaction-cost measures, and evaluate stochastic RL methods throughout ten independent seed repetitions.
8. **Transfer Dynamics Analysis:** Examine peer acceptance, gradient alignment, sharing frequency, and realised performance to identify associations between knowledge transfer behaviour and GARL improvement or deterioration.

1.6 Project Scope

This study examines daily observations of nine US-listed equities from 2001 to 2025. All active trading strategies use the same set of twenty causal technical features. The LSTM, TCN, causal Transformer encoder, and RL agents receive the most recent twenty feature vectors as a temporal window. Random Forest, by contrast, uses the contemporaneous feature vector, while ARIMAX combines the exogenous technical features with its autoregressive and moving average structure.

Forecasting models convert predicted five-day returns into continuous positions bounded between short and long exposure. RL agents directly select incremental actions that decrease, retain, or in-

crease discrete target positions. All Active strategies compute their targets every five trading days and are evaluated under fixed experimental assumptions concerning execution delay, transaction fees, slippage, short-borrow charges and minimum position changes. The passive benchmarks retain their typical buy-and-hold or daily rebalancing rules.

The selected equities provide a controlled multi-stock setting rather than representing the entire US equity market. Consequently, the conclusions apply only to the equities, period, data and execution assumptions examined in this study.

1.7 Dissertation structure

The remainder of the dissertation is organised as follows. Section 2 presents the methodology, including the data, causal validation framework, portfolio construction, model implementations, hyperparameter tuning, and evaluation measures. Section 3 reports performance across the development folds and final holdout, followed by RL seed variation and GARL knowledge transfer diagnostics. Section 4 interprets the findings and reflects on the methodological choices and study boundaries. Section 5 concludes by mentioning the main contribution and direction for future research. The appendices provide supplementary configuration, tuning, and robustness information.

2 Methodology

This chapter outlines the framework adopted to compare passive allocation, forecast-based trading, Single-Agent Reinforcement Learning, and Group-Agent Reinforcement Learning (GARL). The methodology aims to attribute differences in generalisation performance primarily to the learning approaches, rather than to inconsistencies in data, transaction cost assumptions, position limits, or evaluation periods. Hence, all active strategies use the same stock universe, causal information set, decision frequency, portfolio accounting rules, and walk-forward partitions.

The chapter first presents the overall experimental design, data collection and preprocessing procedures. It then defines the common trading and execution framework, followed by descriptions of each baseline and the GARL variants. Finally, the chapter details hyperparameter selection, repeated training, performance evaluation, and measures implemented to guarantee reproducibility.

2.1 Experimental Design

This study adopts a comparative generalisation experiment. The evaluated strategies are divided into three methodological families:

- **Passive allocation strategies:** equal-weight buy-and-hold and daily equal-weight rebalancing.
- **Forecast-first active strategies:** ARIMAX, Random Forest, LSTM, TCN, and Transformer Encoder models.
- **Direct sequential decision strategies:** joint and independent-organised A2C, PPO, and DDQN agents, together with GARL-DDAL and selective GARL-DDAL.

These families represent passive allocation, prediction followed by position construction, and direct policy learning. All active strategies share the same information set, decision schedule, exposure constraints and execution model.

The primary unit of evaluation is the daily return of an equal-capital portfolio consisting of nine stocks. Each stock is treated as an individual portfolio sleeve with identical initial capital allocation. Net returns at the sleeve level are calculated after deducting trading costs and short-borrow charges. The overall portfolio return is computed as the arithmetic mean of the sleeve-level net returns.

$$r_{p,t}^{\text{net}} = \frac{1}{9} \sum_{i=1}^9 r_{i,t}^{\text{net}} \quad (5)$$

where $r_{i,t}^{\text{net}}$ is the net return of the stock sleeve i on trading day t . The equation defines an equal-allocation portfolio sleeve aggregation convention, in which each stock-specific strategy contributes one-ninth of the portfolio return on each trading day. This approach prevents any single stock from dominating the active strategy comparison due to historical capital appreciation. However, averaging sleeve returns daily implicitly restores equal economic importance across sleeves, without explicitly modelling or accounting for capital transfers between them. Consequently, this method cannot be interpreted as nine initially equal, fully self-financing sub-portfolios with capital weights that drift over time without reallocation. This accounting convention is applied consistently to all active strategies, but its difference from the explicitly modelled passive allocation rules is acknowledged as a limitation.

All active trading strategies make scheduled decisions once on a five-day trading cadence. This decision frequency matches the five-day forecasting horizon and curtails excess turnover driven by high-frequency market noise. To support consistent comparison, every model-generated position passes through an identical execution pipeline that enforces uniform exposure boundaries, a no-trade

threshold to filter negligible adjustments, standard execution latency, and a consistent transaction cost model.

The experiment includes five development outer folds followed by a final holdout period beginning in January 2023. Development folds are used for model comparison, model optimisation and diagnostic analysis. The final holdout is kept outside these decisions and is evaluated only after the experimental specification has been fixed.

Passive allocation strategies are deterministic for a given evaluation period. Reinforcement learning methods are stochastic because their outcomes are influenced by parameter initialisation, action sampling, and optimisation trajectories. To address this variation, each RL baseline is trained and evaluated using ten independent seeds, and its performance is reported using the mean and dispersion across those repetitions. The neural forecasting models are also affected by random initialisation and mini-batch ordering, but computational constraints permit only one fixed seed for each LSTM, TCN and Transformer configuration. As a result, the reported variability in RL results reflects training-seed variation on a common market path, whereas uncertainty from neural-forecasting initialisation is not quantified. This difference is treated as a limitation when interpreting comparisons between model families.

2.2 Data and Stock Universe

2.2.1 Stock selection

The experiment uses the following nine US-listed equities representing three major industries:

Technology: NVIDIA Corp. (NVDA), Apple Inc. (AAPL), Microsoft Corp. (MSFT)

Financials: JPMorgan Chase & Co. (JPM), Bank of America Corp. (BAC), Morgan Stanley (MS)

Industrials: Caterpillar Inc. (CAT), RTX Corp. (RTX), Boeing Co. (BA)

The equity universe was selected purposively to provide three related but distinct trading tasks within each of the technology, financial and industrial sectors. Securities were required to have sufficiently long and continuous price histories to support the common evaluation period and to exhibit adequate trading activity for a daily-frequency experiment. Stocks within the same sector may share common market drivers, whereas stocks from different sectors may demonstrate weaker or regime-dependent relationships. This heterogeneous structure is relevant for assessing whether knowledge exchanged between GARL agents is advantageous or leads to negative transfer. The selected stock universe is not mean to represent the entire US equity market. Instead, it provides a controlled multi-stock environment in which all models can be evaluated consistently over a shared historical period.

Consequently, the conclusions are limited to the securities, time frame, and execution assumptions examined.

2.2.2 Data Collection and Cleaning

Daily open, high, low, close, and volume (OHLCV) data were retrieved from Yahoo Finance using the `yfinance` interface. The requested collection period from 1 January 2001 to 31 December 2025. Adjusted price fields are used to account for corporate actions, including stock splits and distributions. The downloaded time series are sorted chronologically, and their time-zone information is standardised before feature construction. Technical indicators are calculated independently for each stock, using only data available on or before each observation date.

A common trading calendar is required for all models. Consequently, the final dataset uses only the intersection of dates for which every stock has a valid closing price and a complete feature vector. This technique prevents changes in portfolio composition due to stock-specific missing observations. Since the longest indicator uses a 200-day moving window, the usable aligned sample begins after the necessary warm-up period rather than on the initial download date. No synthetic observations or market scenarios are introduced. Missing trading days are not forward-filled, as they typically correspond to exchange holidays.

2.3 Feature Engineering and Prediction Target

2.3.1 Causal technical features

Let $P_{i,t}$ denote the adjusted closing price of stock i on trading day t . Each feature vector $\mathbf{x}_{i,t}$ is computed from information available on or before day t . Rolling statistics are calculated using backward-looking windows ending at t , while exponentially weighted statistics are updated recursively based on historical observations. Centred rolling windows and future price data are not utilised. The feature set comprises 20 variables covering return, trend, momentum, volatility, price range, and volume information.

Category	Feature	Construction and Interpretation
Returns	1-, 5-, and 10-day returns	Percentage price changes over short horizons.
Momentum	20-day rate of change	Relative price change over the preceding 20 trading days.
Trend	SMA ratios	Closing price divided by its 10-, 30-, and 200-day simple moving averages, minus one.
Trend	EMA ratios	Closing price relative to its 12- and 26-day exponential moving averages.
Momentum	MACD and signal	Normalised difference between 12- and 26-day exponential averages, and its nine-period signal line.
Momentum	RSI	14-day relative strength index, rescaled for numerical stability.
Trend strength	ADX	14-day average directional index measuring non-directional trend strength.
Volatility	10- and 30-day volatility	Rolling standard deviation of daily simple returns.
Volatility	Bollinger-band score	Standardised displacement of price from its 20-day moving average.
Price range	ATR ratio	14-day average true range normalised by the closing price.
Price range	Range position	Position of the current close within its 20-day closing price range.
Volume	Volume score	Standardised volume over a 20-day rolling window, clipped to restrict extreme spikes.
Volume	OBV slope	Recent change in on-balance volume over a 10-day rolling window.

Table 1. Technical features used by active trading strategies.

These indicators are not presumed to have stable predictive power individually. They provide a common information set from which linear, tree-based, sequence-learning, and RL methods may extract distinct relationships. Using the same underlying features also reduces the likelihood that any single method gains an advantage solely because it has access to more comprehensive market information.

2.3.2 Five-day return target

The forecast-based models estimate the forward five trading day return:

$$y_{i,t}^{(5)} = \frac{P_{i,t+5}}{P_{i,t}} - 1 \quad (6)$$

The target is consistent with the five-day decision interval employed in the trading simulation. Compared with next-day returns, the five-day target captures longer price movements and reduces the influence of isolated daily noise. Nevertheless, it is short enough to reflect an active trading horizon.

For models that require sequential observations, the state at date t contains the latest 20 feature vectors:

$$\mathbf{X}_{i,t} = [\mathbf{x}_{i,t-19}, \dots, \mathbf{x}_{i,t}] \in \mathbb{R}^{20 \times 20} \quad (7)$$

The twenty-day window corresponds to one trading month. It provides recent temporal context without introducing greater computational cost and estimation variance associated with very long sequences. The LSTM, TCN, and Transformer encoder process this matrix, and their architectures determine how they aggregate temporal information.

Standardisation parameters for neural sequence models and RL agents are estimated exclusively from the training sample. Validation and test observations are transformed using these parameters. The return target for neural forecasting models is also standardised within the training set and then converted back to return units before portfolio construction. Tree-based and statistical baselines retain their respective preprocessing methods rather than undergoing the same numerical transformation.

2.4 Walk-Forward Validation and Leakage Prevention

2.4.1 Outer Development Folds

Random train-test splitting is unsuitable for financial time series, as it may train on data that postdate the evaluation period. Therefore, this experiment employs chronological walk-forward validation.

Five outer folds are constructed within the development period. Each fold consists of a training interval followed by a non-overlapping test interval. The first fitted training sample contains 1255 observations, corresponding to approximately five years. As the experiment progresses, newly available observations are incorporated into training, and the training history is limited to 1864 observations (about seven and a half years) to prevent models from being overly influenced by outdated market relationships.

For outer fold k , let T_k and E_k denote its training and evaluation date sets, respectively. Strict chronological separation requires $\max(T_k) < \min(E_k)$. A five-observation purge is additionally applied at each boundary because the forward return target depends on prices over the subsequent five trading days. Fig. 3 presents the resulting walk-forward structure.

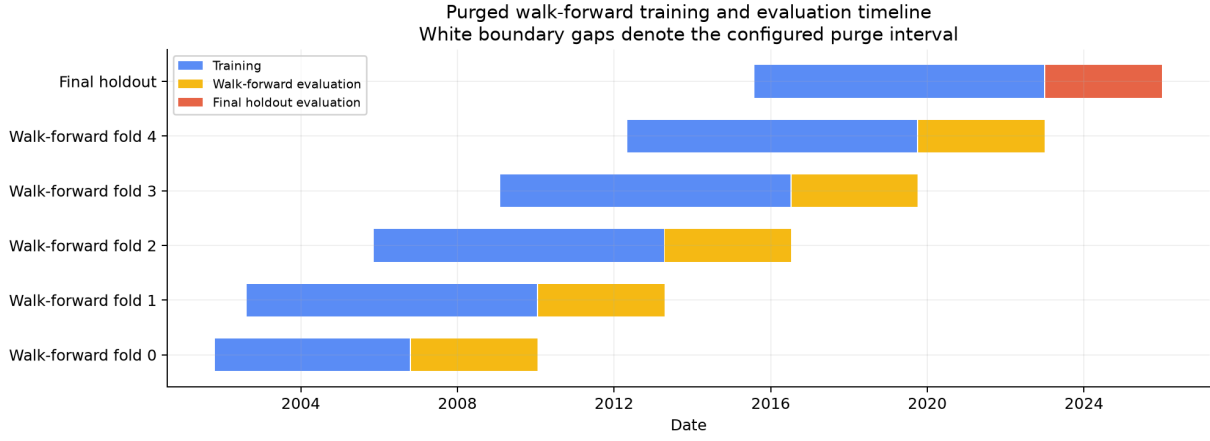


Fig. 3. The timeline of the walk-forward outer folds

2.4.2 Purging at split boundaries

The five trading day forward return defined in equation (6) uses prices observed after the feature date. Therefore, a training observation located immediately before an evaluation boundary could have a target whose calculation overlaps the evaluation period. To prevent this form of label leakage, a five-observation purge is applied between every training and evaluation interval. If t_{test} denotes the first evaluation observation, the final admissible training feature date must satisfy $t + 5 < t_{\text{test}}$. The same boundary treatment applies to the outer folds, internal forecasting-validation folds, RL validation split, and final holdout period.

2.4.3 Nested hyperparameter tuning

Forecast-based models are tuned using three chronological inner folds inside each outer training fold. Candidate hyperparameters are evaluated on these internal validation intervals and the selected specification is then refitted on the outer-training data before evaluation on the outer test interval.

RL optimisation is more computationally expensive because each control variable requires interaction with a simulated trading environment. Hence, RL parameters are assessed using a final validation segment taken from the outer training set. Approximately the first 80% of the period is used for policy estimation and the remaining 20% for validation, with the same five-day purge at the boundary.

For both model families, candidate configurations are ranked by the net Sharpe ratio obtained on the relevant validation period. This ratio is calculated from cost-adjusted daily returns using a zero risk-free rate. It balances average realised return against volatility and therefore provides a more suitable model selection objective than cumulative return alone. Other metrics such as net cumulative return, drawdown and other risk measures are also retained for performance reporting.

2.4.4 Final holdout evaluation

The final holdout period begins on the first available trading day in January 2023 and continues until the end of the sample in December 2025. This period is excluded from selecting or adjusting indicators, model weights, thresholds, learning rates, and other model settings.

Prior to holdout evaluation, each model is refitted using only the training observations that precede the holdout period. Following this refitting, the strategy is assessed on the previously unseen holdout data. This separation allows estimation of performance on a consistent, out-of-sample market period after all principal parameter choices have been finalised.

2.5 Portfolio Construction and Execution Framework

The learning models produce different forms of output. Forecast-based models estimate a continuous five-day return, whereas RL agents select a discrete action that changes their target exposure. To make their realised performance comparable, all active strategies pass their outputs through the same position and execution framework. This framework defines how model outputs become target positions, when those targets may be executed, how positions drift between decisions, and how trading frictions are deducted. The passive benchmarks retain their defining allocation rules but use the same price data and proportional trading cost assumptions.

2.5.1 Passive allocation benchmarks

Two passive strategies serve as baseline reference points for interpreting the economic value of active forecasting and policy-learning methods. These two strategies do not rely on external market signals or predictive inputs.

For nine equities in the selected universe, the equal-weight buy-and-hold strategy assigns an initial portfolio weight $w_{i,t_e}^{\text{BH}} = \frac{1}{N}$ to each stock. After this allocation, the number of shares remains fixed and no target rebalancing is performed. Portfolio weights subsequently evolve according to the relative price performance of the constituent stocks. Price-driven weight drift does not generate turnover, so the strategy causes proportional trading costs only at the time of the initial allocation.

The daily equal-weight rebalanced strategy restores $w_{i,t}^{\text{EW}} = \frac{1}{N}$ at the beginning of each trading interval. Let $\tilde{w}_{i,t}$ denote the weight of stock i immediately before rebalancing, following the preceding price movement. Daily rebalancing turnover is defined as $\tau_t^{\text{EW}} = \sum_{i=1}^N \left| \frac{1}{N} - \tilde{w}_{i,t} \right|$. This strategy preserves equal diversification but causes repeated costs from correcting price-driven weight changes.

Comparison with the buy-and-hold strategy distinguishes the effect of maintaining equal weights from the effect of allowing successful stocks to become increasingly influential.

To maintain a consistent return convention, both passive portfolios begin with zero exposure on the first observation of each evaluation interval and establish their equal-weight allocation on the subsequent observation. The initial purchase causes the same transaction fee and slippage rates applied elsewhere in the experiment. Both strategies remain long-only and target zero cash exposure after entry.

2.5.2 Forecast-to-position conversion

The ARIMAX, Random Forest, LSTM, TCN, and Causal Transformer encoder models estimate the forward five trading day return $\hat{r}_{i,t}^{(5)}$ for each asset i . Each forecast is mapped to a bounded continuous target exposure $z_{i,t}^* \in [-1, 1]$ via a clipped mean-variance allocation rule [26]:

$$z_{i,t}^* = \text{clip} \left(\frac{\hat{r}_{i,t}^{(5)}}{\lambda \hat{\sigma}_{i,\text{train}}^2}, -1, 1 \right) \quad (8)$$

where $\lambda = 10$ is a fixed risk-aversion coefficient, and $\hat{\sigma}_{i,\text{train}}^2$ is the unconditional sample variance of five-day training targets for asset i . To prevent lookahead leakage, $\hat{\sigma}_{i,\text{train}}^2$ is computed strictly on the training partition and held constant across the out-of-sample evaluation interval.

Equation (8) corresponds to the single-asset constrained solution of a mean-variance objective, scaling the position size inversely with the historical target variance. Because this transformation is applied invariantly across all forecasting architectures, observed performance differentials reflect the quality and calibration of the underlying predictions rather than position sizing artifacts. This rule isolates single-asset directional conviction and deliberately excludes cross-asset covariance estimation and dynamic conditional volatility modelling.

2.5.3 Reinforcement learning target positions

Unlike forecast-based models, reinforcement learning agents do not produce return forecasts. Rather, each learned policy directly selects an incremental action $a_{i,t} \in \{-1, 0, +1\}$, which corresponds to a decrease, retention, or increase in the target exposure level within the discrete position set $\mathcal{Z} = \{-1, 0, +1\}$.

Given the previous target $z_{i,t-1}^*$, the updated target exposure is defined as:

$$z_{i,t}^* = \text{clip}(z_{i,t-1}^* + a_{i,t}, -1, 1) \quad (9)$$

An action $a_{i,t} = 0$ retains the previous target. At the next scheduled decision, rebalancing occurs only if the difference between that target and the drifted pre-trade exposure exceeds the no-trade threshold. Between scheduled decisions, exposure is allowed to drift unless forced clipping is required to maintain the exposure bounds. A reversal from $+1$ to -1 requires two decrease actions at successive scheduled decisions, with zero exposure during the intervening five-day interval.

2.5.4 Execution timing and trading frictions

A target generated from information available on day t is shifted forward by one observation before execution, preventing the strategy from earning a return that occurred before its decision became available. Active targets are reconsidered every five trading days. On a scheduled decision date, a trade is executed only when the difference between the proposed target and the price-drifted pre-trade exposure exceeds a threshold of 0.20. Between decisions, exposure is allowed to drift, except when forced deleveraging is required to preserve the bound $[-1, 1]$.

Trading incurs a transaction fee of five basis points and slippage of two basis points per unit of absolute exposure change. Short exposure additionally incurs an annualised borrow charge of 50 basis points, allocated across 252 trading days. For held exposure $z_{i,t}$, pre-trade exposure $\tilde{z}_{i,t}$, asset return $r_{i,t}$, proportional trading cost $c = 0.0007$ and annual borrow rate $b = 0.005$, the daily net sleeve return is:

$$r_{i,t}^{\text{net}} = \max \left\{ -1, z_{i,t} r_{i,t} - c |z_{i,t} - \tilde{z}_{i,t}| - \frac{b}{252} \max(-z_{i,t}, 0) \right\} \quad (10)$$

The lower bound imposes limited liability at sleeve level. Portfolio equity begins at \$100,000 and compounds from the mean of the nine sleeve returns. Cash earns no interest. During RL training, the turnover component of c is multiplied by two to discourage policies that exploit noisy short-term fluctuations through excessive trading. Reported backtests always use the unmodified seven basis points trading cost.

2.6 Model Implementations

2.6.1 Forecasting baselines

Each forecasting model is fitted independently for each stock. The static ARIMAX model is trained once on the entire outer training partition and kept fixed throughout the evaluation interval. The rolling ARIMAX model employs a sliding historical window and periodically re-estimates its parameters using causally revealed evaluation observations, incorporating five-day targets that have fully elapsed. Candidate ARIMAX orders are restricted to $p, q \in \{0, 1, 2\}$ and $d \in \{0, 1\}$. Rolling ARIMAX window sizes of 126, 252, and 504 observations, as well as refitting intervals of 5, 10, and 20 observations, are evaluated.

The Random Forest is fitted to each contemporaneous technical feature vector. The hyperparameter search varies the number of trees from 150 to 450, maximum tree depth from 3 to 9, and minimum leaf size from 10 to 50. Feature subsampling is performed using the square root of the total number of available features.

The neural predictors use training-only feature standardisation and standardise five-day targets using only training data, then transform predictions back to the original return scale. The LSTM architecture consists of two recurrent layers followed by a non-linear regression head. The TCN contains four residual blocks with dilations of 1, 2, 4, and 8; each block applies two weight-normalised causal convolutions, ReLU activations, and dropout. The Transformer encoder projects features into a learnt latent space, adds learnt positional embeddings, and applies two causally masked encoder layers with four attention heads. All three neural predictors use a dropout rate of 0.2, mean squared error loss, Adam optimisation, a batch size of 64, and gradient clipping at unit norm. Hidden layer width, learning rate, and number of training epochs are selected within each outer training fold.

Fig. 4 summarises the different input representations and architectural biases of the forecasting models while highlighting their common prediction target and position-construction rule.

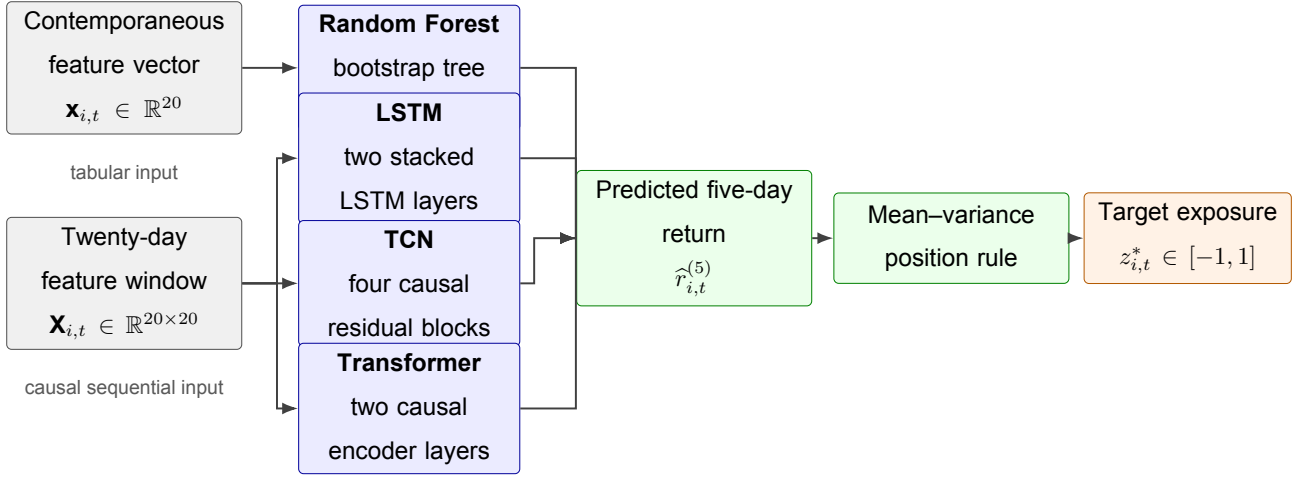


Fig. 4. Forecasting architectures and common position-construction pipeline. Random Forest uses a contemporaneous feature vector, whereas the deep-sequence models process a twenty-day feature window. All four models predict the same five-day return target and apply the common mean–variance position rule.

2.6.2 Reinforcement learning state representation

For each asset i , the RL state space is constructed by flattening the standardised 20×20 temporal feature matrix and appending the current target exposure, resulting in a 401-dimensional input vector. All RL agents use the same causal TCN architectural backbone for feature extraction, and each maintains a fully decoupled set of learnable parameters. This backbone consists of four residual convolutional blocks, each with 32 channels, a kernel size of $k = 3$, and dilation factors $d \in \{1, 2, 4, 8\}$, which provide an effective receptive field covering the entire 20-day lookback window. The extracted temporal feature representation is concatenated with the current target exposure and passed through a two-layer fully connected network with tanh activation functions. Encoder dropout is disabled by default.

Each environment transition encompasses a five-trading-day execution horizon and generates the compounded net return realised over that period. Since the base discount factor $\gamma = 0.95$ is specified at a daily resolution, the value function and advantage targets are scaled across the multi-day decision step using the effective transition discount $\gamma^5 = 0.95^5$.

2.6.3 RL algorithms and agent organisation

A2C, PPO, and DDQN share an identical temporal feature encoder, state observation space, and incremental action formulation. The A2C policy uses Generalised Advantage Estimation (GAE) [27] with $\lambda_{\text{GAE}} = 0.95$ to balance bias and variance in the estimated advantages. An entropy regularisation term promotes exploration and the critic loss is weighted by a coefficient of 0.5. PPO uses the same advantage estimator and performs four optimisation epochs per rollout trajectory with the clipped

surrogate objective. The DDQN implementation uses a uniform experience replay buffer of 5,000 transitions, a batch size of 64, Huber loss minimisation, and separate online and target networks. Exploration is governed by an ϵ -greedy schedule decaying from 1.0 to 0.05. Consistent with DDQN methodology, the online network selects the greedy action, while the target network evaluates its Q-value.

For each algorithm, the centralised joint organisation uses one shared temporal representation and asset-specific discrete action heads, optimising the mean reward across the nine individual stock environments. In contrast, the independent organisation instantiates a model, an optimiser, and an environment per asset, prohibiting cross-asset parameter sharing. During evaluation, stochastic exploration is deactivated, and actor-critic agents execute the action with the highest probability, while DDQN selects the action that maximises the predicted Q-value.

2.6.4 GARL-DDAL and selective knowledge transfer

Both GARL variants instantiate an individual A2C agent for each equity asset, initialised with identical parameters from a common template. Thereafter, they maintain independent model parameters, optimisers, environments, random streams, and locally asynchronous clocks. The independent A2C baseline uses the identical parameter initialisation but executes throughout training without peer knowledge transfer.

Each GARL agent performs an initial warm-up phase consisting of 30 private training epochs. Thereafter, local gradient computation continues, and peer gradient assimilation occurs every two local epochs. Asynchrony is modelled via an event-driven simulation in which agents progress according to heterogeneous, seed-controlled inter-update intervals. During a sharing event, a receiving agent aggregates the most recent gradient updates from up to three peer agents.

Fig. 5 distinguishes continuous local optimisation from the less frequent peer transfer events and shows where the standard and selective GARL-DDAL mechanisms diverge.

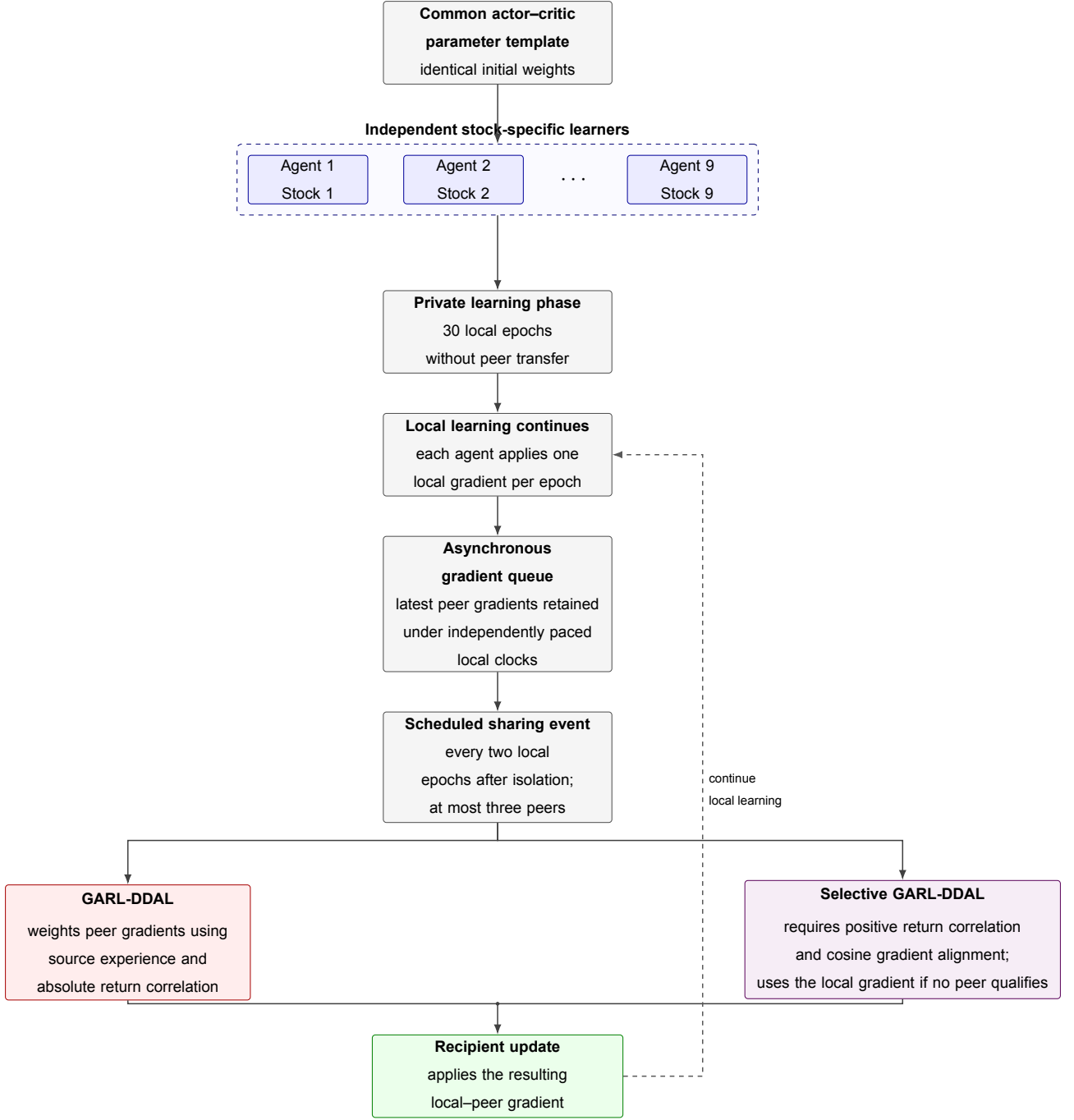


Fig. 5. Training and asynchronous knowledge transfer workflow of the financial GARL-DDAL variants. After common initialisation and thirty private epochs, each stock-specific agent continues local optimisation. At sharing events scheduled every two post-isolation local epochs, standard GARL-DDAL weights queued peer gradients using source experience and absolute return correlation, whereas Selective GARL-DDAL additionally requires positive return correlation and cosine gradient alignment.

Standard GARL-DDAL measures task similarity using the absolute correlation of training returns. According to equation (4), local and peer gradients are weighted equally between normalised source experience and task relevance. In contrast, Selective GARL-DDAL requires both a strictly positive return correlation and a directional gradient alignment that exceeds a specified cosine similarity threshold. For peers meeting these criteria, final blending weights combine the DDAL experience-relevance score with directional alignment, and the aggregated peer gradient is blended equally with the re-

ceiver’s local gradient. If no peer meets the criteria, the agent performs an isolated local update. Peer selection decisions, gradient cosine similarities, queue sizes, and communication events are recorded for subsequent transfer analysis.

2.7 Training and Hyperparameter Selection

The supervised forecasting models are tuned using three chronological inner validation folds nested within each outer training partition. Fifteen hyperparameter configurations are produced using Optuna [28] and evaluated according to their mean inner-validation net Sharpe ratio. The configuration with the highest mean validation score is refitted using the complete outer training partition.

Given the computational cost of repeated environment simulation, RL baselines are tuned using a grid of nine candidate configurations, evaluated on the final 20% of the outer training period. A five-day purging window separates policy training from validation to prevent lookahead leakage. Each tuning run consists of 20 epochs under a fixed random seed of 42. The parameter search space includes learning rate and entropy regularisation for A2C, clipping threshold for PPO, exploration decay rates and target-network update frequencies for DDQN, and cosine alignment thresholds for Selective GARL-DDAL. The top-performing configuration for each algorithm and outer fold is then fixed across ten final repetitions to avoid seed-specific overfitting.

Final RL policy training is conducted for a fixed budget of 100 epochs, with 32 transitions per roll-out trajectory. Early stopping and checkpoint restoration are disabled to ensure that all policies are evaluated exclusively on their terminal model parameters. Statistical variation is assessed across ten stochastic repetitions using random seeds 42 to 51, while deterministic and supervised forecasting baselines are evaluated using only seed 42.

2.8 Performance Evaluation

The primary evaluation measures are net cumulative return, annualised arithmetic return, Compound Annual Growth Rate (CAGR), annualised volatility, annualised Sharpe ratio, and maximum drawdown. Following Sharpe [29], the annualised Sharpe ratio is calculated from daily net portfolio returns as

$$S_{\text{annual}} = \sqrt{252} \cdot \frac{\bar{r}_{\text{net}} - r_{f,d}}{\hat{\sigma}(r_{\text{net}})} \quad (11)$$

where $\bar{r}_{\text{net}}$ and $\hat{\sigma}(r_{\text{net}})$ denote the sample mean and standard deviation of daily net returns. The daily risk-free rate $r_{f,d}$ is fixed at zero, consistent with the experiment assumption that the uninvested

cash earns no interest. The same convention is applied to every strategy and evaluation period. The remaining primary measures are defined as

$$\begin{aligned}
R_{\text{cum}} &= \frac{V_T}{V_0} - 1 = \prod_{t=1}^T (1 + r_{p,t}^{\text{net}}) - 1, \\
R_{\text{ann}} &= 252 \cdot \bar{r}_p^{\text{net}}, \\
\text{CAGR} &= \left(\frac{V_T}{V_0} \right)^{252/T} - 1, \\
\sigma_{\text{ann}} &= \sqrt{252} \cdot s \left(r_{p,t}^{\text{net}} \right), \\
\text{MDD} &= \min_{1 \leq t \leq T} \left(\frac{V_t}{\max_{0 \leq u \leq t} V_u} - 1 \right)
\end{aligned} \tag{12}$$

where T is the number of daily return observations and V_t is portfolio equity obtained by compounding the daily net portfolio returns. Annualised arithmetic return scales the mean daily return, whereas CAGR measures the constant annual growth rate implied by beginning and terminal equity. Maximum drawdown is reported as a negative percentage relative to the preceding equity peak.

Secondary metrics assess risk-adjusted asymmetries, trading frictions, and tail risk, including the Sortino and Calmar ratios, downside deviation, portfolio turnover, gross and net exposures, transaction cost drag, win rate, profit factor, 95% Value at Risk (VaR), 95% Conditional Value at Risk (CVaR), return skewness, excess kurtosis, and the Ulcer Index. For RL models, performance measures are summarised across ten independent random seeds using their mean, dispersion, and 95% confidence intervals.

3 Results

This chapter reports the experimental results. Evidence from the five development folds is presented first to assess temporal consistency, followed by the previously unseen final holdout evaluation. Reinforcement learning results are reported as means across ten independently seeded training runs, whereas each non-RL baseline is represented by one fitted run per evaluation period.

3.1 Development Fold Performance

Strategy	Mean Sharpe	Sharpe SD	Cumulative return (%)	Max drawdown (%)
Equal-weight rebalancing	0.791	0.478	64.6	-38.8
GARL-DDAL	0.777	0.439	55.9	-34.7
Independent A2C	0.776	0.416	54.1	-32.5
Selective GARL-DDAL	0.776	0.437	54.5	-33.7
Buy-and-hold	0.765	0.485	62.8	-38.5
Independent PPO	0.739	0.412	46.5	-28.7
Random Forest	0.558	0.116	15.9	-16.8
Joint A2C	0.409	0.572	20.2	-19.8
Joint PPO	0.377	0.316	13.2	-21.4
Static ARIMAX	0.239	0.177	9.3	-23.9
Independent DDQN	0.167	0.141	5.7	-18.7
LSTM	0.073	0.369	3.4	-19.6
Transformer	0.062	0.277	1.3	-19.5
Joint DDQN	0.048	0.155	2.6	-23.5
TCN	0.048	0.282	3.7	-18.9
Rolling ARIMAX	-0.041	0.213	-3.8	-23.8

Notes: Repeated-model results are first averaged within each fold and then aggregated across the five development folds. Sharpe SD measures variation between fold-level means.

Table 2. Mean development fold performance across five walk-forward folds.

Table 2 summarises performance across the five chronological development folds. Daily equal-weight rebalancing achieved the highest mean Sharpe ratio of 0.791, followed closely by GARL-DDAL (0.777), independent A2C (0.776), Selective GARL-DDAL (0.776), and buy-and-hold (0.765). Their mean Sharpe ratios were similar despite differences in agent organisation and trading rules. GARL-DDAL produced a mean cumulative return of 55.9%, compared with 54.1% for independent A2C and 54.5% for the selective variant.

Performance varied materially between periods. For example, GARL-DDAL's fold-level Sharpe ratio ranged from 0.35 in the first fold to 1.452 in the fourth. Independent A2C followed a similar pattern, ranging from 0.379 to 1.452. The two passive strategies also performed most strongly in the third and fourth folds. Random Forest was the strongest forecasting baseline on average, with a mean Sharpe ratio of 0.558, while the remaining statistical and sequence models exhibited weaker and less consistent results. Rolling ARIMAX produced a negative mean Sharpe ratio of -0.041.



Fig. 6. Sharpe ratio stability for each strategy across the five walk-forward development folds. RL values represent means across ten independently seeded repetitions within each fold.

Fig. 6 shows that the leading passive and actor-critic strategies responded similarly to changing market conditions. Their Sharpe ratios increased during the third and fourth folds.

3.2 Final Holdout Performance

Strategy	Cumulative return (%)	Annual return (%)	Annual volatility (%)	Sharpe	Max drawdown (%)	Daily turnover (%)
Random Forest	28.4	8.5	4.1	2.059	-5.1	2.04
Equal-weight rebalancing	166.2	34.6	18.8	1.840	-22.6	1.13
Independent A2C	128.0	28.7	15.7	1.835	-19.3	0.12
GARL-DDAL	124.9	27.9	15.2	1.833	-18.4	0.13
Independent PPO	124.2	28.1	15.5	1.810	-19.1	0.11
Selective GARL-DDAL	116.9	26.5	14.6	1.809	-17.9	0.12
Buy-and-hold	228.1	43.0	25.0	1.716	-27.6	0.13
Joint PPO	42.7	11.4	10.5	1.043	-15.6	2.09
LSTM	17.4	5.7	8.0	0.710	-7.5	6.46
Joint A2C	34.3	8.4	10.1	0.704	-17.1	1.50
Independent DDQN	19.3	6.2	9.3	0.652	-13.8	8.48
TCN	9.0	3.2	7.7	0.412	-7.5	9.71
Transformer	0.6	0.6	8.9	0.066	-11.2	8.73
Joint DDQN	0.8	0.4	8.9	0.055	-17.9	7.53
Static ARIMAX	-1.4	0.2	11.9	0.017	-11.2	6.13
Rolling ARIMAX	-2.1	-0.4	7.3	-0.059	-20.7	11.70

Notes: RL results are means across ten independently seeded repetitions. Passive and forecast-first strategies are represented by one fitted run. Performance is measured after transaction costs.

Table 3. Final holdout performance comparison across all baselines.

Table 3 reports the holdout results from January 2023 to December 2025. Random Forest achieved the highest holdout Sharpe ratio of 2.059, with an annualised volatility of 4.12%. However, its net cumulative return of 28.4% was lower than those of the passive benchmarks and top-performing actor-critic strategies. This profile was characterised by mean gross exposure of 21.7%, consistent with both its low volatility and more moderate capital growth. By comparison, daily equal-weight rebalancing generated a cumulative return of 166.2% and a Sharpe ratio of 1.84 while maintaining approximately full market exposure.

Independent A2C was the highest-ranked RL strategy by mean Sharpe ratio, producing 1.835 and a cumulative return of 128%. GARL-DDAL followed closely with a Sharpe ratio of 1.833 and a cumulative return of 124.9%. Independent PPO achieved 1.81 and 124.2%, while Selective GARL-DDAL achieved 1.809 and 116.9%. Equal-weight buy-and-hold yielded the highest cumulative return of 228.1%, although its higher volatility of 25% reduced its Sharpe ratio to 1.716. Its maximum drawdown of -27.6% was also larger than those of independent A2C (-19.3%), GARL-DDAL (-18.4%), and Selective GARL-DDAL (-17.9%).

This risk-return difference was accompanied by contrasting levels of market exposure. Mean gross exposure was 79.3% for independent A2C, 78% for GARL-DDAL, 79% for independent PPO, and 73.7% for Selective GARL-DDAL, compared with approximately 100% for both passive strategies.

Relative to buy-and-hold, the leading RL strategies therefore generated lower terminal returns while maintaining lower exposure and experiencing smaller maximum drawdowns.

Performance separated more clearly beyond the leading stock-specific actor-critic strategies. Joint PPO achieved a Sharpe ratio of 1.043, while LSTM, TCN, and the encoder-only Transformer achieved 0.71, 0.412, and 0.066, respectively. Static and rolling ARIMAX produced cumulative returns of -1.4% and -2.1%. DDQN also produced lower risk-adjusted performance than the corresponding actor-critic alternatives under both agent organisations.

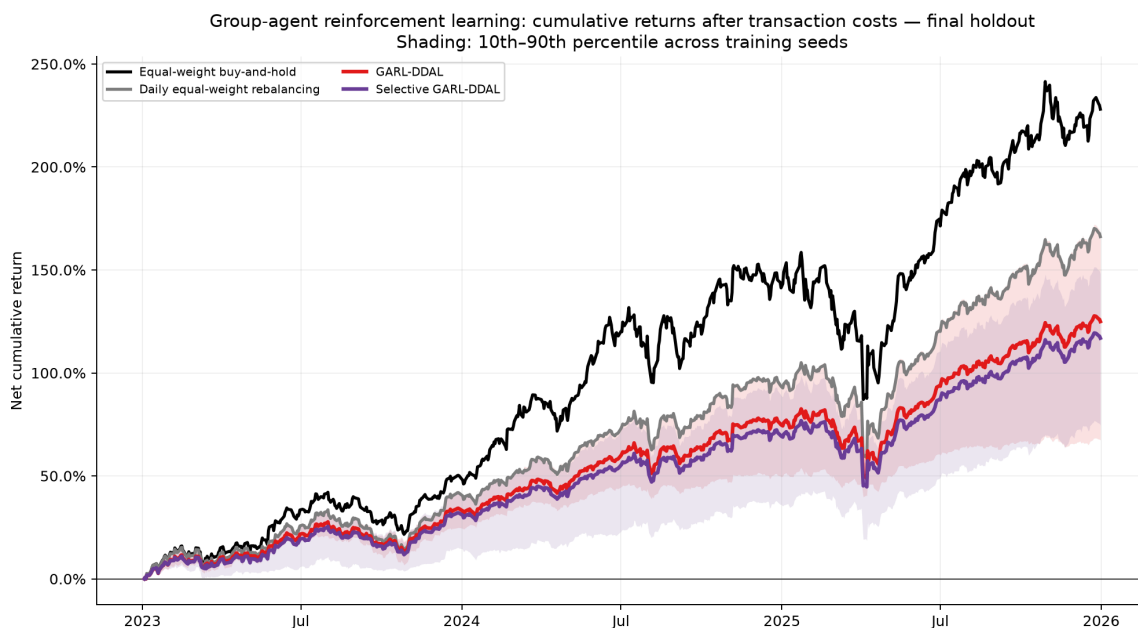


Fig. 7. Net cumulative returns of the GEARL variants and passive strategies during the final holdout period. Shaded regions represent the 10th–90th percentile range across the ten RL training seeds.

Fig. 7 shows that both GEARL variants maintained positive net growth throughout most of the holdout period and recovered following the early 2025 drawdown. Standard GEARL-DDAL finished above its selective counterpart, while the percentile bands quantify variation across stochastic training seeds. The greater terminal wealth of the passive strategies was consistent with their approximately full market exposure, whereas the GEARL variants operated at lower exposure and experienced smaller drawdowns as reported in Table 3.

3.3 RL Seed Variation and Trading Activity

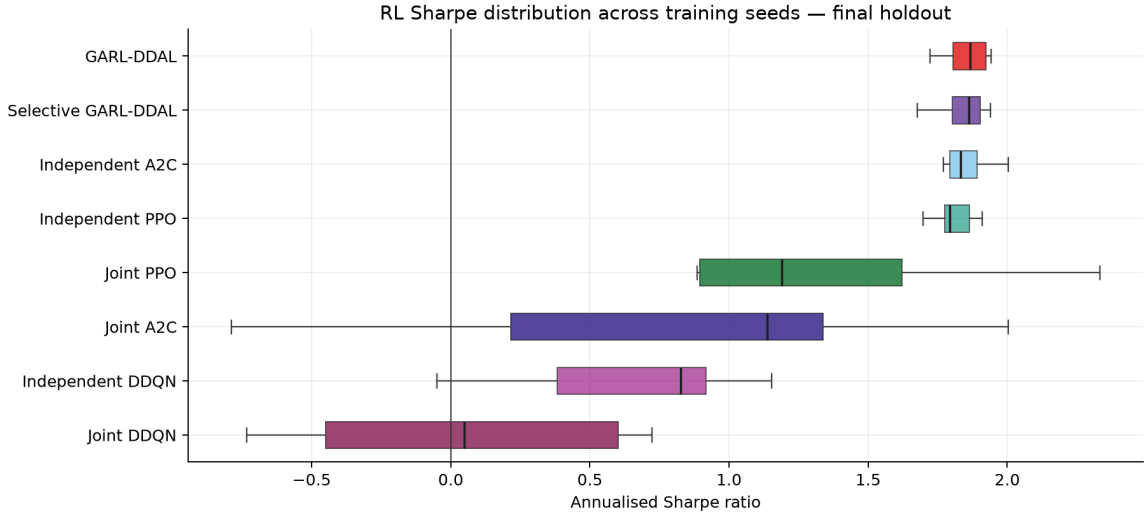


Fig. 8. Distribution of final holdout Sharpe ratios across ten training seeds for each RL strategy.

Fig. 8 shows a clear separation between the leading stock-specific actor-critic policies and the more variable joint-agent and DDQN alternatives. Independent A2C had the narrowest 95% confidence interval half-width among the four leading RL strategies, at 0.079, compared to 0.138 for GARL-DDAL, 0.134 for independent PPO, and 0.11 for Selective GARL-DDAL. The distributions support the mean ordering in Table 3, while their overlap indicates that the differences among the four leading strategies are small relative to seed uncertainty.

The leading independent and group agent policies traded infrequently. Their mean daily turnover ranged from 0.11% to 0.13%, with cumulative transaction cost drag between 0.13% and 0.17%. Independent DDQN exhibited a different trading profile, with daily turnover of 8.48% and cost drag of 5.9%.

3.4 GARL Knowledge Transfer Diagnostics

Given the holdout performance of GARL-DDAL and independent A2C in Table 3, transfer logs were examined to establish how frequently peer information affected local learning. Standard GARL-DDAL incorporated a shared update in 33.4% of recorded holdout agent epochs. Selective GARL-DDAL shared less frequently as intended, in approximately 26.4% of epochs. Across its candidate-sharing events, the selective mechanism accepted 45.8% of available peers, corresponding to an average of 1.27 accepted peer gradients. The mean cosine alignment among accepted gradients was 0.34. Filtering altered the transfer process as intended, but it did not translate into a higher aggregate holdout return or Sharpe ratio than standard GARL-DDAL and independent A2C.

Diagnostic	Development folds	Final holdout
GARL shared-update rate (%)	33.4	33.4
Selective shared-update rate (%)	25.3	26.4
Selective peer-acceptance rate (%)	42.6	45.8
Mean accepted peers	1.17	1.27
Mean accepted alignment	0.334	0.340

Notes: Shared-update rates are calculated over all agent epochs. Peer-acceptance rates and mean accepted peers are calculated over candidate-sharing events. Mean accepted alignment is conditional on at least one peer gradient being accepted.

Table 4. Knowledge transfer diagnostics for the GARL variants.

Transfer diagnostics were also compared with realised performance across individual runs. Among the 50 development fold–seed combinations, the mean accepted gradient alignment did not exhibit a clear monotonic association with the Sharpe ratio (Spearman correlation $\rho = -0.055$). A within-fold rank comparison yielded a similarly small association ($\rho = 0.061$). Although a positive association was observed across the ten holdout seeds ($\rho = 0.794$), this trend was not evident during development periods. These descriptive results indicate that alignment may be informative in specific runs, but do not demonstrate a consistently beneficial relationship between selective transfer and realised performance.

4 Discussion

4.1 Results Interpretation

The results show that portfolio exposure and agent organisation were at least as consequential as model complexity. Buy-and-hold accumulated the greatest terminal wealth during a strongly rising holdout market, consistent with its approximately full market exposure. The leading actor-critic policies maintained lower market exposure and generated lower terminal wealth, but also experienced smaller maximum drawdowns. Random Forest occupied a different point on this trade-off, achieving the highest holdout Sharpe ratio through moderate positive returns combined with low volatility and market exposure.

The contrast between stock-specific and joint policies suggests that task specialisation was valuable. Independent A2C and PPO could adapt their policies to individual equities, whereas each joint agent learnt a common action mapping across heterogeneous assets. This larger state and joint action spaces may have increased the difficulty of credit assignment, contributing to the greater seed disper-

sion observed among the joint policies. The weaker DDQN results are also consistent with off-policy replay becoming less representative as market conditions change. More generally, the observed seed sensitivity supports the existing concerns regarding reproducibility in deep RL [5].

GARL-DDAL [25] preserved stock-specific learning while incorporating asynchronous peer gradients and achieved risk-adjusted performance close to independent A2C and PPO. This outcome demonstrates that knowledge sharing retained competitive aggregate performance, although it did not provide a consistent advantage over isolated learning. This pattern is compatible with cross-asset relationships changing across market regimes [2]. The selective extension reduced sharing and rejected conflicting gradients as intended, but the positive alignment–Sharpe association observed in the hold-out was not reproduced across development folds. Instantaneous gradient agreement was insufficient to identify knowledge that remained economically useful out of sample, motivating transfer measures that account for the persistence and regime dependence of cross-asset relationships.

4.2 Project Reflection and Limitations

The main challenge in this project was to maintain methodological comparability across models with different outputs, training procedures and agent organisations. A common causal data processing pipeline, execution layer and set of portfolio constraints allowed their realised performance to be evaluated under consistent conditions. Repeating every RL assessment across ten seeds also exposed training uncertainty that would have been concealed by a single fitted policy. Together, these design choices prioritised controlled comparison and reproducibility across a diverse set of strategies.

The experiment findings were conditional on nine selected US equities, daily observations and one final holdout period. Neural forecasting models were evaluated using one fixed seed, so their training variation was not quantified. Furthermore, the equal-sleeve aggregation convention restores equal economic importance across active strategy sleeves without explicitly modelling capital transfers, and the zero risk-free rate makes the reported Sharpe ratios relative to a zero return benchmark instead of the contemporaneous cash yield. Future work should thus evaluate broader asset universes and market regimes, repeat stochastic forecasting models, and develop regime-aware GARL relevance measures that assess whether transferred knowledge remains economically useful out of sample.

5 Conclusion

This study established a chronological and cost-aware framework for evaluating asynchronous GARL in stock trading. Stock-specific actor-critic policies produced the strongest risk-adjusted results among

the RL strategies, while GARL-DDAL retained closely comparable performance when incorporating asynchronous peer gradients. The selective extension altered the transfer process by rejecting conflicting updates, although this did not yield a consistent aggregate improvement over standard GARL-DDAL.

The main contribution in this project is a controlled characterisation of collaborative gradient learning across heterogeneous equities, distinguishing the value of stock-specific policy learning from the incremental effect of peer transfer. The findings indicate that effective financial GARL depends not simply on sharing knowledge, but on whether that knowledge remains economically relevant as cross-asset relationships change. Regime-aware relevance measures provide a feasible direction for developing more adaptive group agent trading framework.

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Appendices

A Experimental Configuration

Table 5 summarises the core settings used across the experiment. The complete configuration file and outputs are in the accompanying codebase.

Setting	Value
Stock universe	Nine US-listed equities
Data period	January 2001–December 2025
Final holdout	January 2023–December 2025
Development folds	Five chronological outer folds
Prediction horizon	Five trading days
Temporal lookback	Twenty trading days
Active-strategy decision interval	Five trading days
Position range	$[-1, 1]$
No-trade threshold	0.2 exposure units
Transaction fee	5 basis points per unit traded
Execution slippage	2 basis points per unit traded
Annual short-borrow charge	50 basis points
RL training repetitions	Ten seeds, numbered 42–51
Final RL training budget	100 epochs
RL rollout length	32 transitions

Table 5. Principal experimental and execution settings.

The software environment is created using the supplied setup script, after which the complete experiment and reporting pipeline can be executed as follows:

```
cd MSc_framework
./scripts/setup_env.sh
source .venv/bin/activate
garl-trading run --config configs/default.toml
```

The experiment directory records the resolved configuration, model metrics, portfolio returns, positions, trading costs, tuning outcomes and GARL transfer diagnostics. This allows the tables and figures reported in the dissertation can be reproduced.

B Hyperparameter Selection

Table 6 summarises the model tuning procedure. Forecasting models are selected using three inner validation folds for each outer fold. RL candidates are evaluated on a purged validation segment at the end of each outer training period.

Model family	Principal tuned parameters	Budget	Objective
ARIMAX	AR and MA orders, differencing order, rolling-window length and refitting interval	Bounded search	Net Sharpe ratio
Random Forest	Number of trees, maximum depth and minimum leaf size	15 trials	Mean inner-fold net Sharpe ratio
LSTM, TCN and Transformer	Hidden width, learning rate and training epochs	15 trials	Mean inner-fold net Sharpe ratio
A2C	Learning rate, entropy coefficient and turnover penalty multiplier	9 candidates	Validation net Sharpe ratio
PPO	Learning rate, clipping threshold and turnover penalty multiplier	9 candidates	Validation net Sharpe ratio
DDQN	Learning rate, exploration decay, target update frequency and turnover penalty multiplier	9 candidates	Validation net Sharpe ratio
GARL variants	Learning rate, turnover penalty multiplier and, for selective GARL, alignment threshold	9 candidates	Validation net Sharpe ratio

Table 6. Summary of the hyperparameter-selection procedure.

C Supplementary Robustness Diagnostics

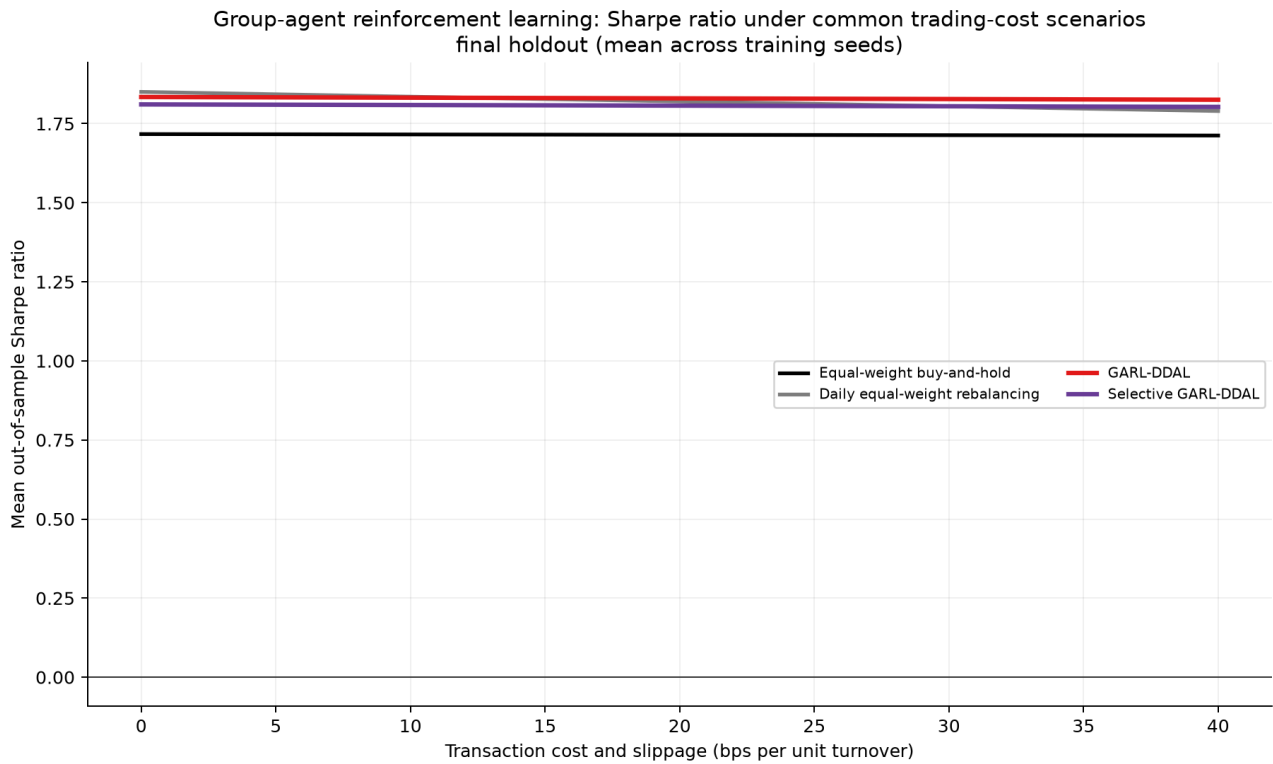


Fig. 9. Transaction-cost sensitivity of GARL-DDAL and Selective GARL-DDAL during the final holdout period.

Fig. 10 presents the equivalent analysis for the independent RL strategies.

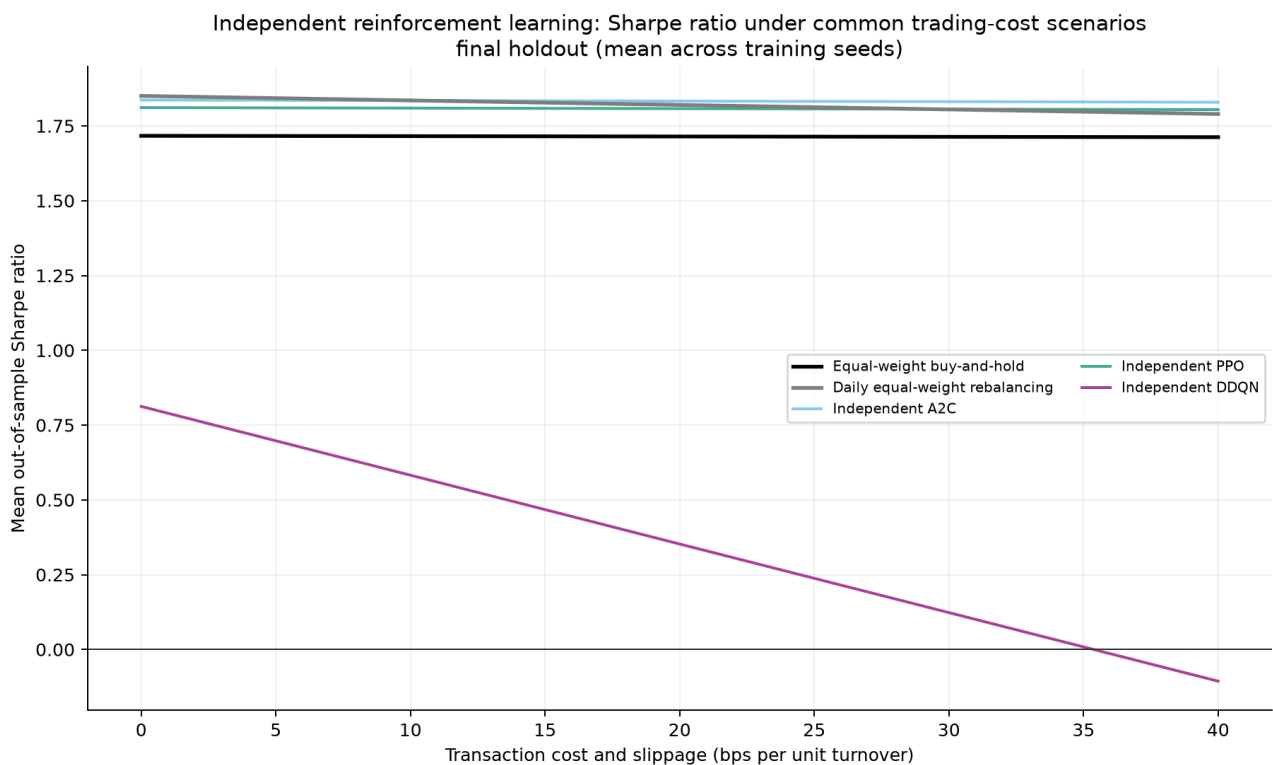


Fig. 10. Transaction-cost sensitivity of the independent RL strategies during the final holdout period.

Training diagnostics are provided in Fig. 11 and Fig. 12. These plots complement the seed-level evaluation in the main report by showing the optimisation behaviour observed over the fixed training budget.

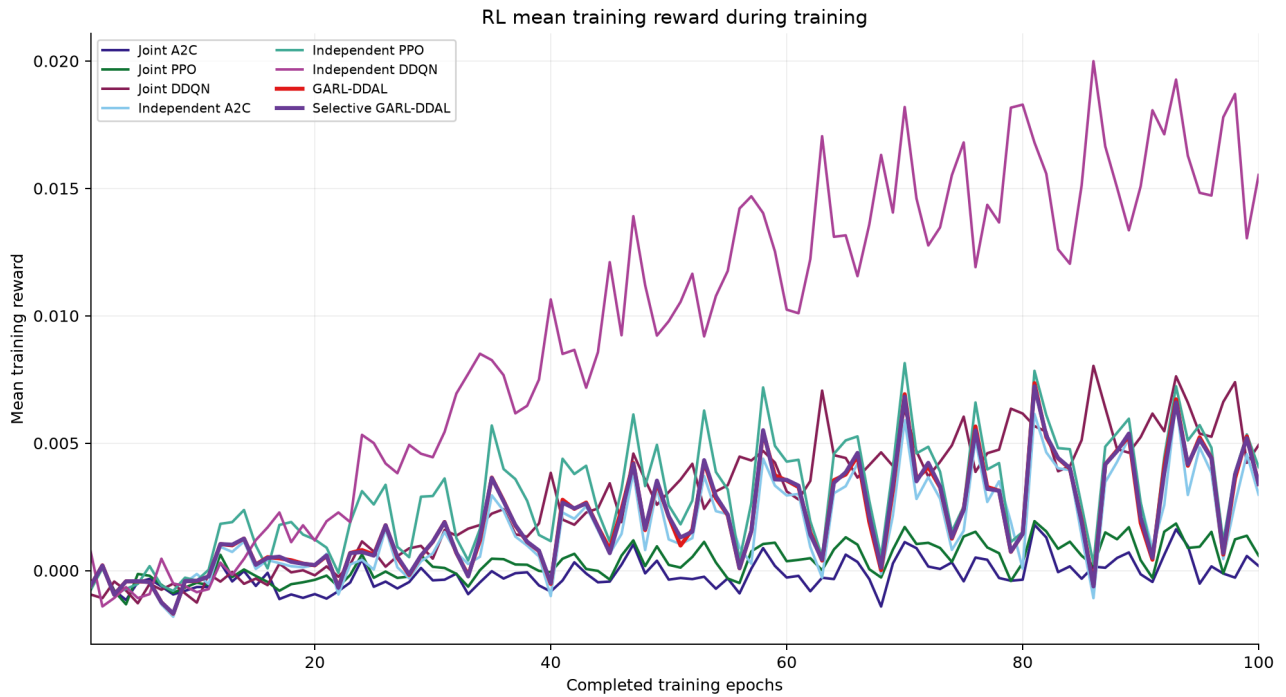


Fig. 11. Training reward trajectories recorded for the RL strategies.

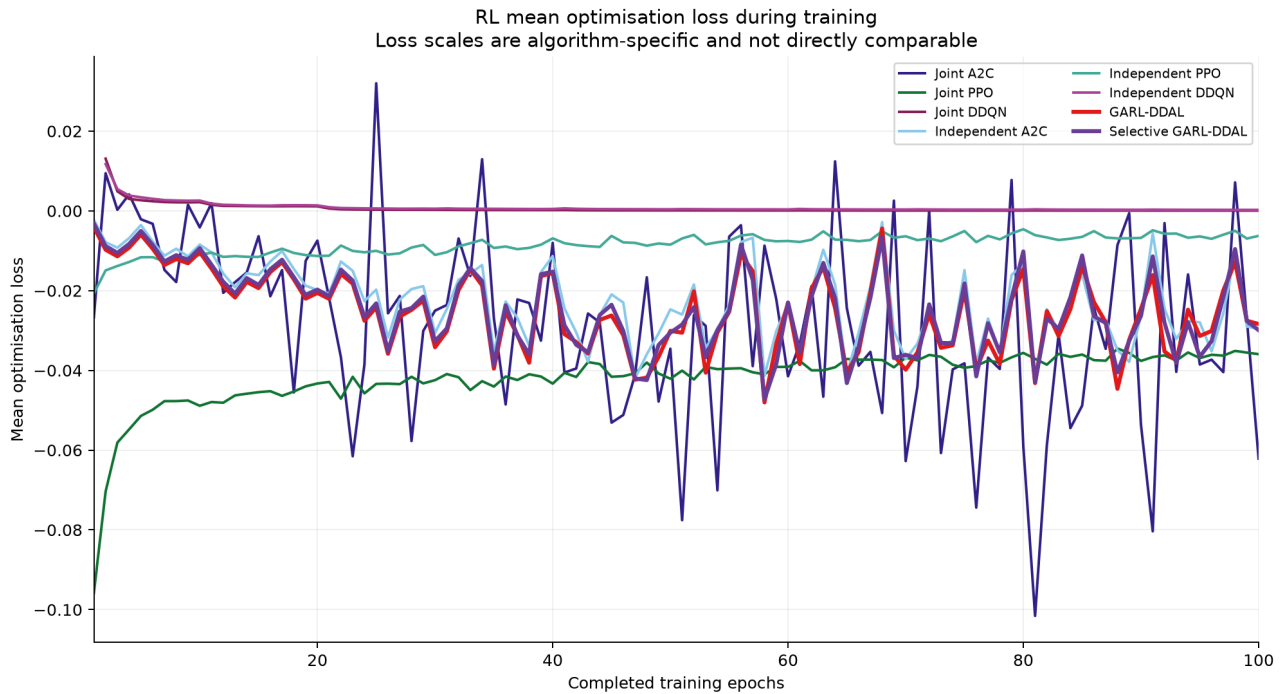


Fig. 12. Training loss trajectories recorded for the RL strategies.